

Citadel

Compiling Enterprise Knowledge

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Enterprise AI is missing a compute layer

Every enterprise system today works on **unstructured documents**.

PDFs

Spreadsheets

Presentations

Emails

Web pages

But none of them are computable.

Every AI system rebuilds the same pipeline

OCR → Parsing → Chunking → Embedding → Retrieval

Each product repeats this independently:

Legal AI

Search

Copilots

Analytics tools

No shared representation exists.

Search is not computation

Search systems:

- Index documents

- Rank text

- Return snippets

They do not understand structure.

They do not execute knowledge.

Search is access. Not computation.

The real problem

Enterprise knowledge has:

- Storage

- Search

- LLMs

But no **execution model**.

Documents are not data structures.

A missing primitive

Software is not executed from source.

It is compiled.

Enterprise knowledge should be too.

Documents are syntax

PDF | DOCX | HTML | Excel | JSON | CSV

All are just **source formats**.

A system should not care about format.

Only structure matters.

Citadel

Citadel compiles enterprise documents into a **canonical knowledge representation**.

Documents → Compiler → Knowledge IR → Execution

One representation. All enterprise knowledge.

Canonical knowledge structure

Document → Sections → Blocks → Tables → Atomic units

Every computation operates on this structure.

This replaces raw document access.

From Retrieval to Execution

Traditional AI:

Question → **Search** → **Text**

Citadel:

Question → **Plan** → **Compute**

Knowledge becomes executable.

The Knowledge Runtime: Amortize and Amortize

Citadel separates:

Compilation

Understand documents

Discover structure

Build representation

Execution

Query knowledge

Compute answers

Preserve evidence

Compile once. Execute infinitely.

AI Models Should Plan. Systems Should Compute.

LLMs are powerful reasoning engines.

But they are not databases.

Citadel uses models for:

- Understanding intent

- Generating plans

- Resolving ambiguity

Citadel runtime performs:

- Retrieval

- Aggregation

- Computation

- Verification

Knowledge Planning replaces Search

Search asks:

"Which document is relevant?"

Citadel asks:

"What computation produces the answer?"

Query → Intent → Evidence → Execution → Answer

Retrieval becomes query planning over knowledge.

Structured Data Emerges Automatically

Documents hide structure.

Citadel discovers:

- Tables

- Relationships

- Schemas

- Hierarchies

Spreadsheets stop being files.

They become databases.

Enterprise documents become queryable systems.

The Execution Loop



Every answer has a computation path.

This System Exists Today

Citadel already compiles:

PDF

|

Office

|

Excel

|

CSV

|

HTML

The Architecture Made Predictions

The thesis predicted:

Any document can become structured knowledge

Different formats can share one representation

Computation can replace retrieval

Provenance can remain exact

Multiple applications can share one substrate

Every prediction became a product capability.

Why This Becomes a Moat

Models improve with data.

Citadel improves with:

- More documents

- More structures

- More schemas

- More queries

The system learns how enterprises organize knowledge.

The moat is the representation layer.

Why Existing Systems Cannot Simply Add This

Current systems assume:

Documents $\rightarrow$ Search $\rightarrow$ Answer

Citadel changes the assumption:

Documents $\rightarrow$ Compilation $\rightarrow$ Computation

This is an architectural shift.

The Fourth Enterprise Data Layer

Computing evolved through abstractions:

Databases → Warehouses → Search → **Citadel**

A canonical computational layer for enterprise knowledge.

The missing primitive between documents and intelligence.

Every programming language needed a compiler.

Every enterprise database needed SQL.

Every document will need a computational representation.

Citadel is building it.